

Amino Acids, Peptides, and Proteins

Organic Chemistry II

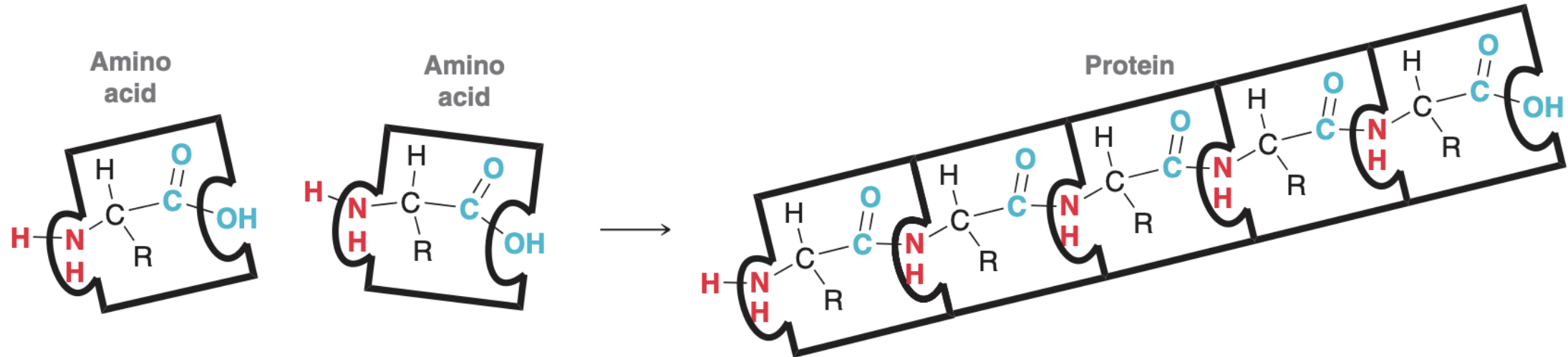
CHEM 2312

Dr. Michael Evans

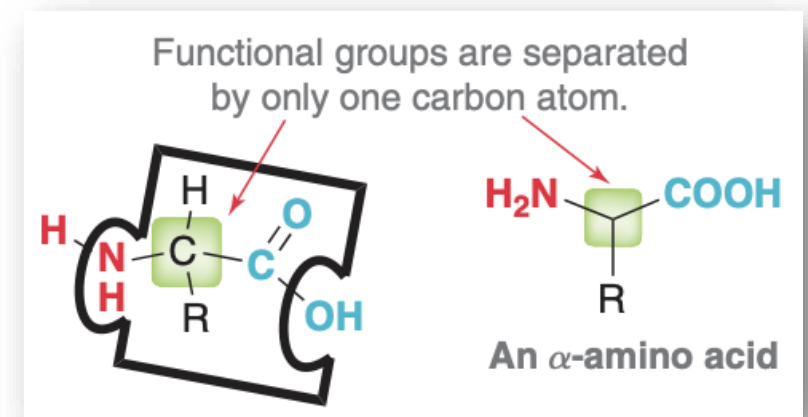
1. Recognize proteins and polypeptides as polymers of α -amino acids.
2. Draw the structure of a general α -amino acid, including stereochemical information.
3. Classify a given amino acid according to its structure and properties.
4. Draw the dominant ionization state of an amino acid in aqueous solution at a given pH, given pK_a values for the amino acid.
5. Determine the isoelectric point of an amino acid given pK_a values.
6. Apply the Hell-Volhard-Zelinsky reaction, amidomalonate synthesis, and Strecker synthesis for the preparation of amino acids with arbitrary structure.
7. Apply the Edman degradation and peptidase enzymes to sequence a polypeptide chain.
8. Design the synthesis of a polypeptide chain with arbitrary primary sequence using protection, coupling, and deprotection reactions.
9. Recognize and interpret the primary, secondary, tertiary, and quaternary structures of proteins.

25.1 Introduction to Amino Acids, Peptides, and Proteins

- **Proteins** are polymers of monomers known as **amino acids**
- Amino acid monomers are linked through amide groups (**peptide bonds**)
- A contiguous chain of linked amino acids is called a **polypeptide**

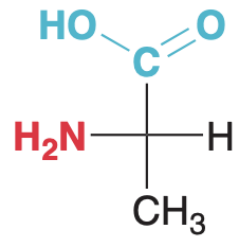


- The most common and important amino acids are **α -amino acids**
- The R group linked to the α -carbon is called the **side chain**

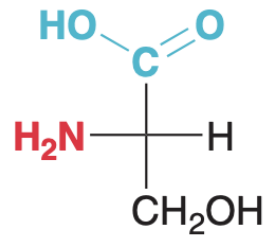


25.2 Naturally Occurring Amino Acids | 1 of 2

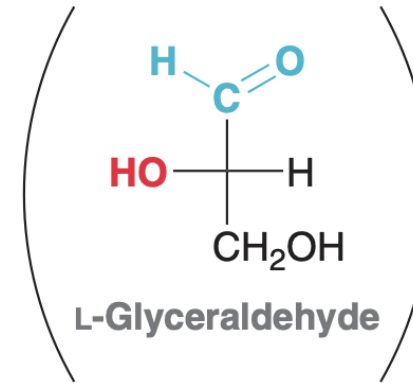
- The α -carbon is very often stereogenic; naturally occurring amino acids have the L configuration
- With the carboxyl group “up” and the side chain “down,” the amino group is on the “Left”



L-Alanine

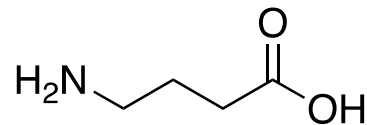


L-Serine

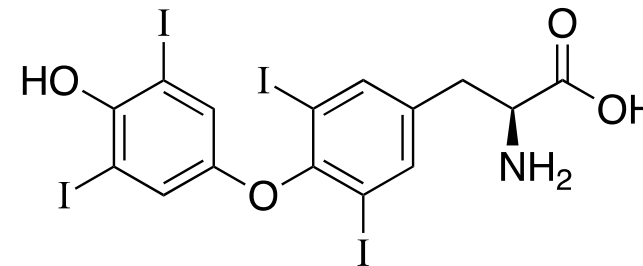


L-Glyceraldehyde

- A few naturally occurring amino acids are not found in proteins but have other functions



γ -aminobutyric acid



thyroxine

25.2 Naturally Occurring Amino Acids | 2 of 2

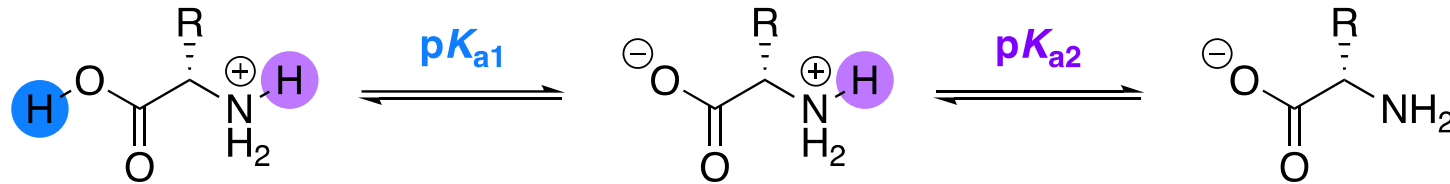
Chart Key: ● ALIPHATIC ● AROMATIC ● ACIDIC ● BASIC ● HYDROXYLIC ● SULFUR-CONTAINING ● AMIDIC ○ NON-ESSENTIAL ○ ESSENTIAL

Chemical Structure single letter code NAME A three letter code DNA codons	ALANINE A <i>Ala</i> GCT, GCC, GCA, GCG	GLYCINE G <i>Gly</i> GGT, GGC, GGA, GGG	ISOLEUCINE I <i>Ile</i> ATT, ATC, ATA	LEUCINE L <i>Leu</i> CTT, CTC, CTA, CTG, TTA, TTG	PROLINE P <i>Pro</i> CCT, CCC, CCA, CCG	VALINE V <i>Val</i> GTT, GTC, GTA, GTG
PHENYLALANINE F <i>Phe</i> TTT, TTC	TRYPTOPHAN W <i>Trp</i> TGG	TYROSINE Y <i>Tyr</i> TAT, TAC	ASPARTIC ACID D <i>Asp</i> GAT, GAC	GLUTAMIC ACID E <i>Glu</i> GAA, GAG	ARGININE R <i>Arg</i> CGT, CGC, CGA, CGG, AGA, AGG	HISTIDINE H <i>His</i> CAT, CAC
LYSINE K <i>Lys</i> AAA, AAG	SERINE S <i>Ser</i> TCT, TCC, TCA, TCG, AGT, AGC	THREONINE T <i>Thr</i> ACT, ACC, ACA, ACG	CYSTEINE C <i>Cys</i> TGT, TGC	METHIONINE M <i>Met</i> ATG	ASPARAGINE N <i>Asn</i> AAT, AAC	GLUTAMINE Q <i>Gln</i> CAA, CAG

Note: This chart only shows those amino acids for which the human genetic code directly codes for. Selenocysteine is often referred to as the 21st amino acid, but is encoded in a special manner. In some cases, distinguishing between asparagine/aspartic acid and glutamine/glutamic acid is difficult. In these cases, the codes asx (B) and glx (Z) are respectively used.

25.2 Acid-base Properties of the Amino Acids

- Amino acids are *never* found in an ionization state without formal charges
 - The carboxyl group is **acidic**
 - The amino group is **basic**
 - The side chain may be acidic, basic, or neutral!
- Major ionization state depends on the pH of the medium in accordance with the [Henderson-Hasselbalch relation](#) (you already know this from Chemical Principles II!)
 - Below pK_{a1} , both the $-\text{CO}_2\text{H}$ group and $-\text{NH}_3^+$ group are protonated
 - Between pK_{a1} and pK_{a2} , the $-\text{CO}_2\text{H}$ group is deprotonated and the $-\text{NH}_3^+$ group is protonated (**zwitterion**)
 - Above pK_{a2} , both the $-\text{CO}_2\text{H}$ group and $-\text{NH}_3^+$ group are deprotonated



25.2 Acidity Constants of the Amino Acids

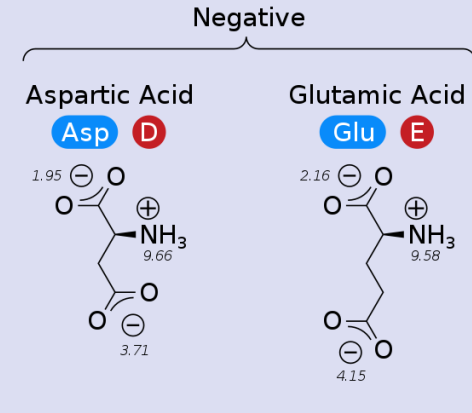
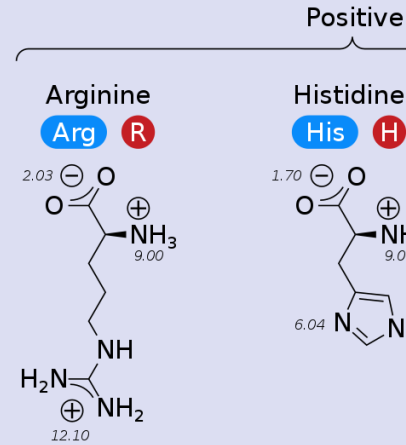
TWENTY-ONE PROTEINOGENIC α -AMINO ACIDS

Side chain charge at physiological pH 7.4

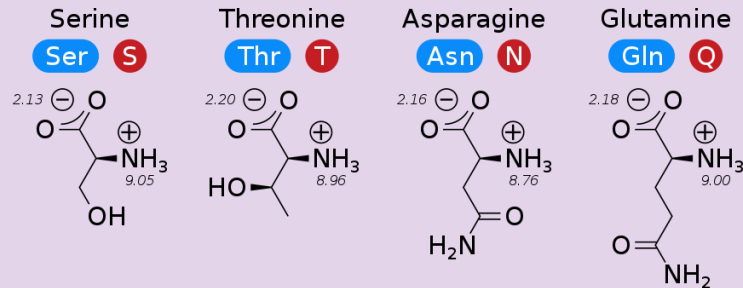
pK_a values shown italicized

⊕ Positive
⊖ Negative

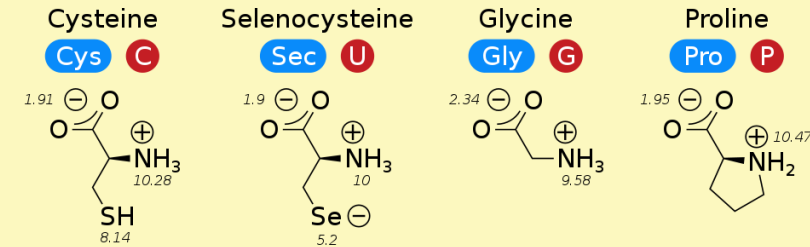
A. Amino Acids with Electrically Charged Side Chains



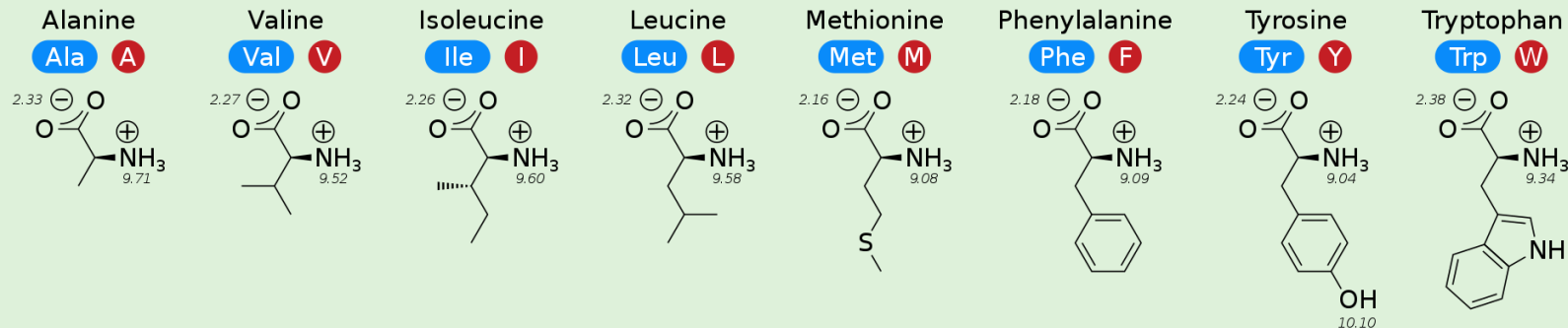
B. Amino Acids with Polar Uncharged Side Chains



C. Special Cases

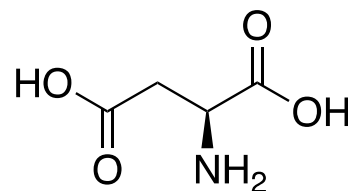
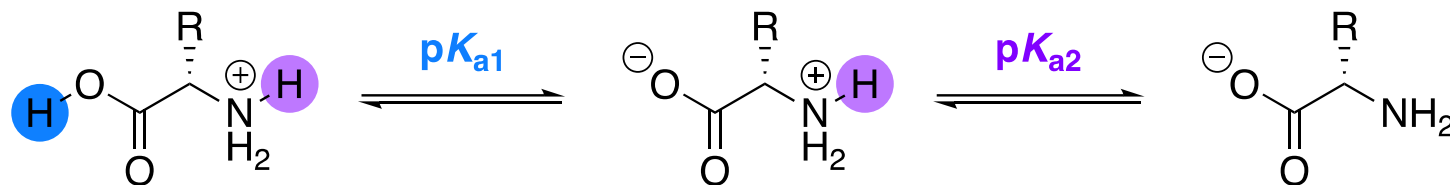


D. Amino Acids with Hydrophobic Side Chains



25.2 Acidic Amino Acids Have Side Chains with Low pK_a

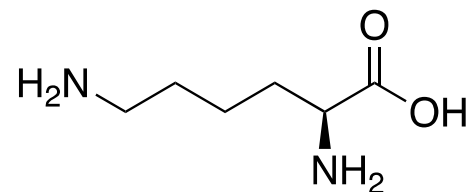
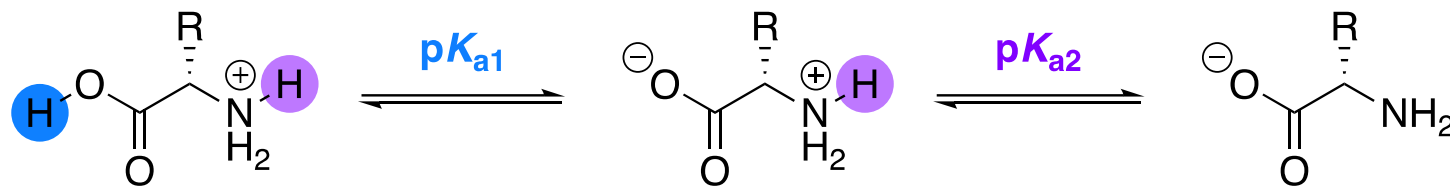
Example. Draw all possible ionization states of the amino acid aspartic acid, ordering them from lowest to highest pH at which they dominate. Relevant pK_a values are 1.95, 9.66, and 3.71.



aspartic acid

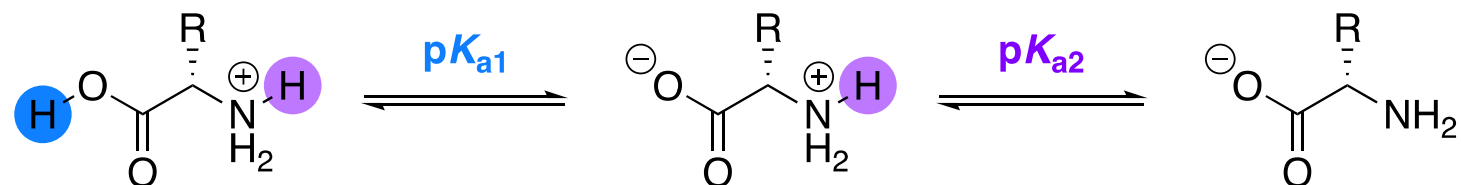
25.2 Basic Amino Acids Have Side Chains with High pK_a

Example. Draw all possible ionization states of the amino acid lysine, ordering them from lowest to highest pH at which they dominate. Relevant pK_a values are 2.15, 9.16, and 10.67.



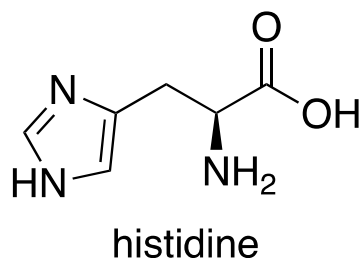
lysine

- The **isoelectric point** is the pH at which the concentration of the zwitterionic form is at its maximum value

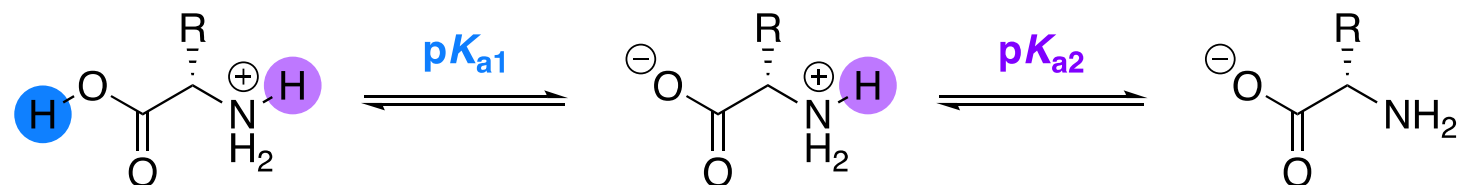


- The isoelectric point is an average of two pK_a values
 - For amino acids with neutral side chains, isoelectric $pH = \frac{1}{2}(pK_{a1} + pK_{a2})$
 - For amino acids with acidic side chains, isoelectric $pH = \frac{1}{2}(pK_{a1} + pK_{a,SC})$
 - For amino acids with basic side chains, isoelectric $pH = \frac{1}{2}(pK_{a2} + pK_{a,SC})$

Example. Determine the isoelectric point of histidine. Relevant pK_a values are 1.70, 9.09, and 6.04.

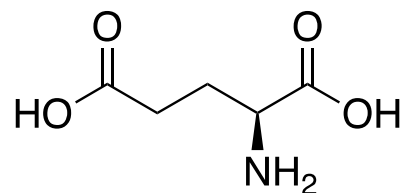


- The **isoelectric point** is the pH at which the concentration of the zwitterionic form is at its maximum value



- The isoelectric point is an average of two pK_a values
 - For amino acids with neutral side chains, isoelectric $pH = \frac{1}{2}(pK_{a1} + pK_{a2})$
 - For amino acids with acidic side chains, isoelectric $pH = \frac{1}{2}(pK_{a1} + pK_{a,SC})$
 - For amino acids with basic side chains, isoelectric $pH = \frac{1}{2}(pK_{a2} + pK_{a,SC})$

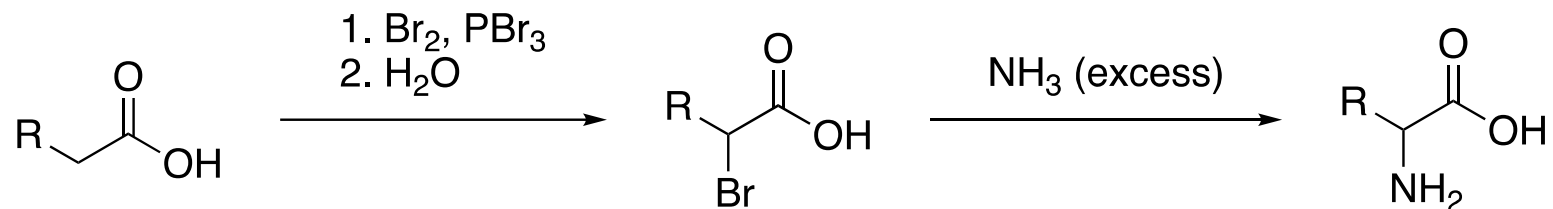
Example. Determine the isoelectric point of glutamic acid. Relevant pK_a values are 2.16, 9.58, and 4.15.



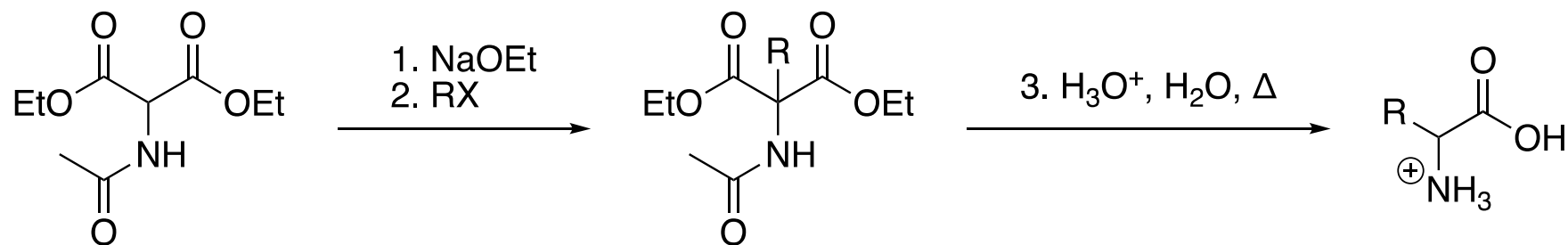
glutamic acid

25.3 Preparations of Amino Acids

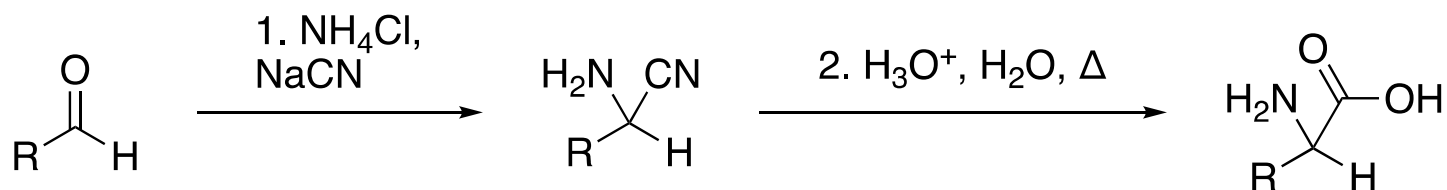
- The **Hell-Volhard-Zelinsky reaction** can be used to prepare α -bromo acids that react with ammonia to furnish amino acids



- The **amidomalonate synthesis** is a variant of the malonic ester synthesis that employs diethyl acetamidomalonate as the starting material

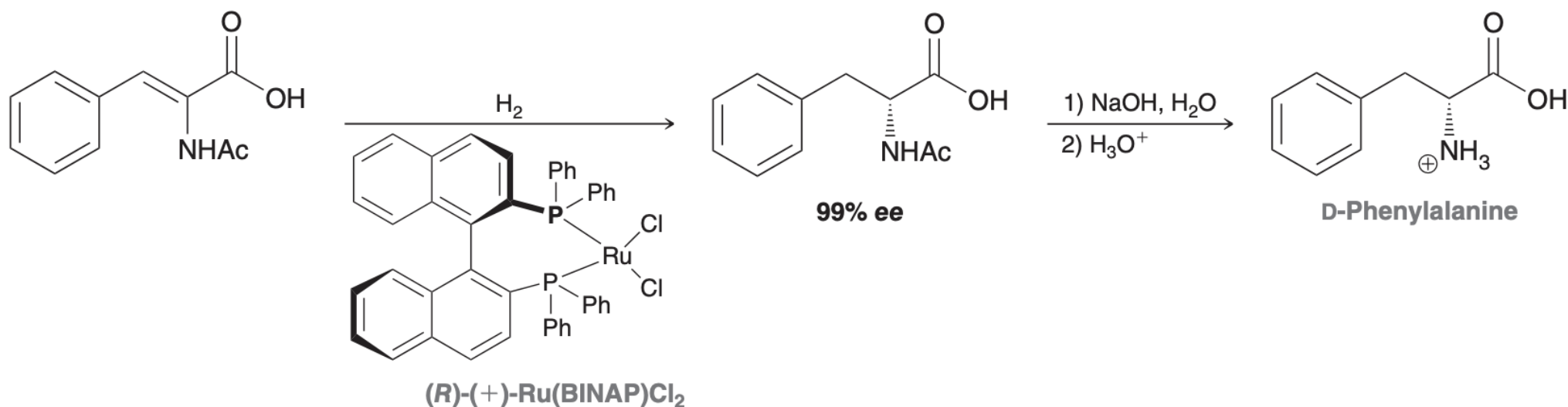


- The **Strecker synthesis** involves imine formation and *in situ* hydrocyanation followed by hydrolysis of the resulting α -amino nitrile



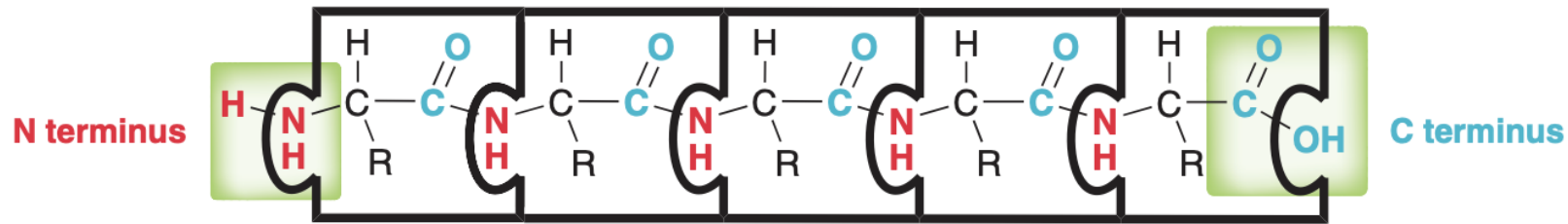
25.3 Enantioselective Synthesis of Amino Acids

- Enantioselective hydrogenation of unsaturated acids with chiral complexes of transition metals is a commonly applied method for the preparation of amino acids in enantioenriched form
- The chiral catalyst creates a bias for one enantiomer over the other—the bias is often extreme!

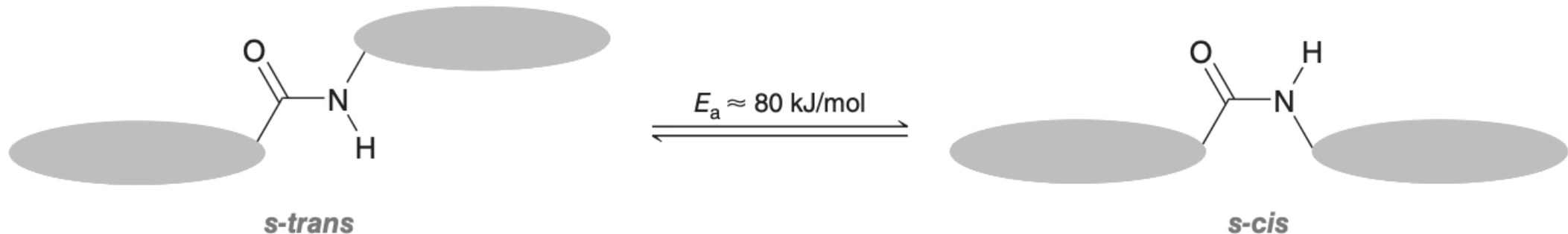


25.4 Geometry of Peptide (Amide) Bonds

- Polypeptides contain a backbone with repeated amide units with an *N*-terminus (amino terminus) and *C*-terminus (carboxyl terminus)
- Polypeptide sequences are listed from the *N*-terminus to the *C*-terminus

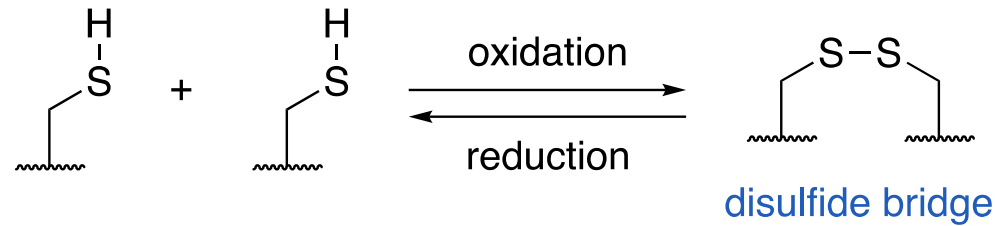


- Due to significant double-bond character in the $\text{C}-\text{N}$ bond, rotation around this bond is severely restricted

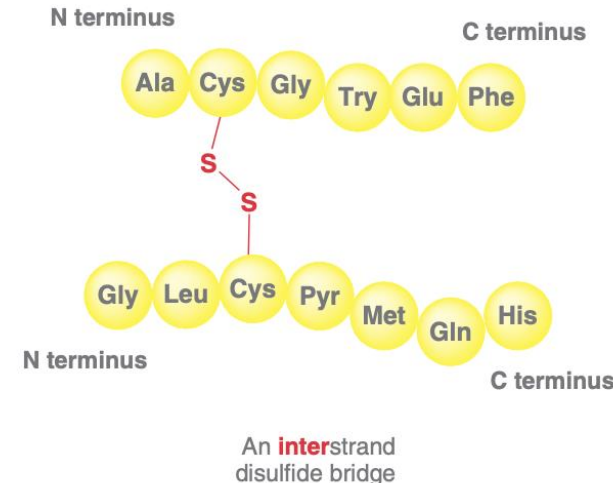
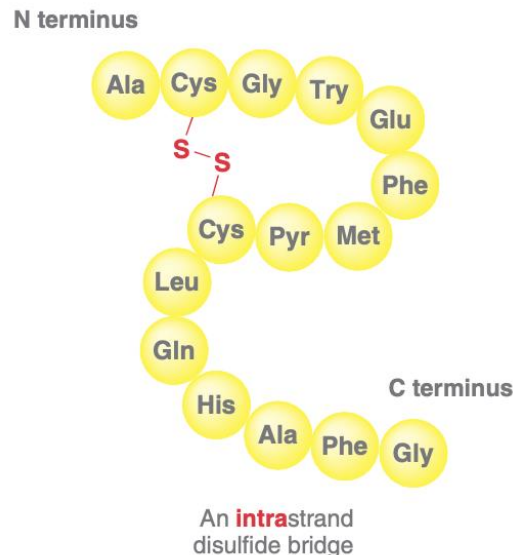


25.4 Disulfide Bonds Confer Structural Rigidity in Polypeptides

- Sulfur atoms in cysteine residues can bond to each other to create a covalent **disulfide bridge**
- Conversion of two thiols to a disulfide (or *vice versa*) is a redox process

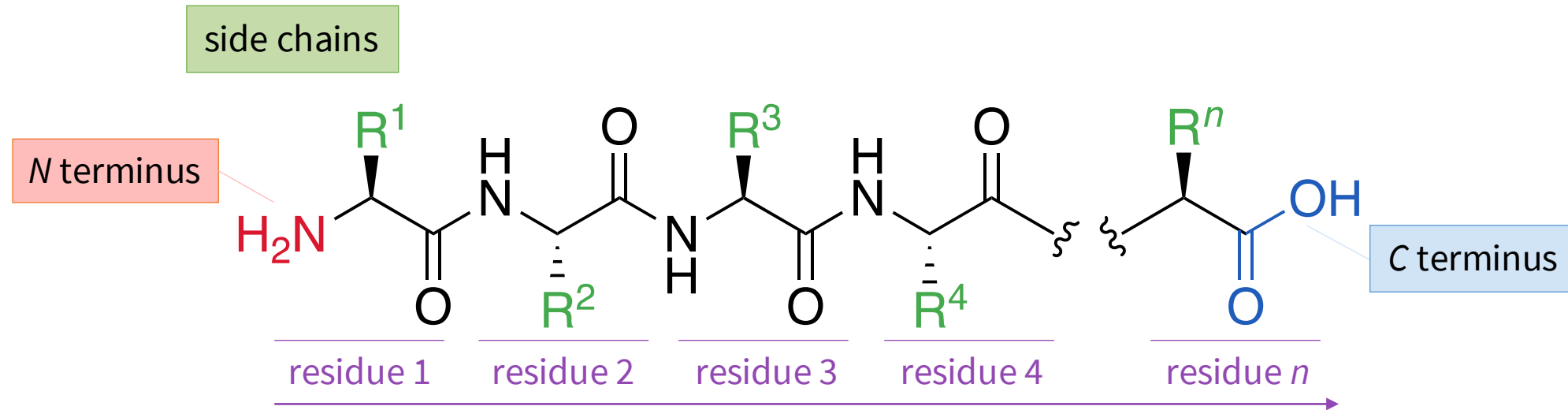


- Disulfide bridges may be introduced via molecular oxidizing agents or protein disulfide isomerase (PDI) enzymes
- May link two residues within a single polypeptide chain or across two chains



25.5 The Problem of Peptide Sequencing

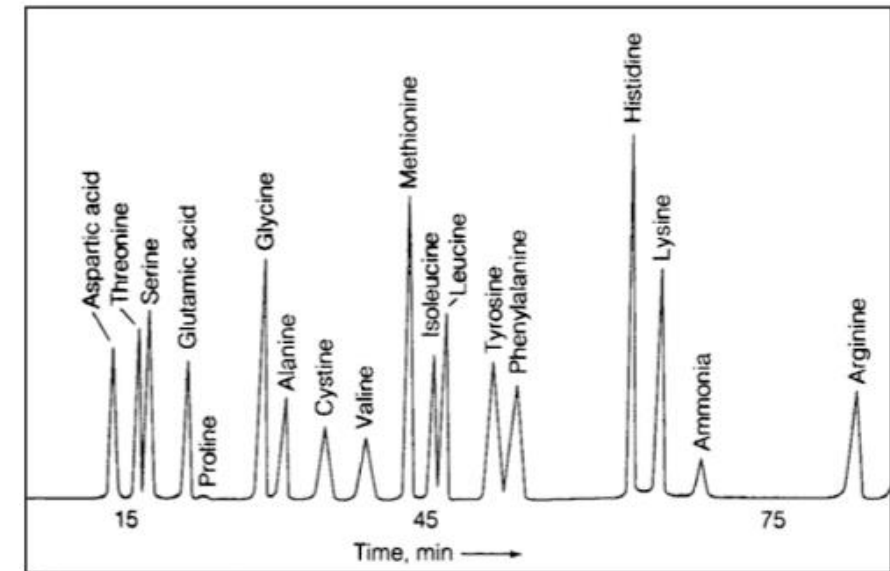
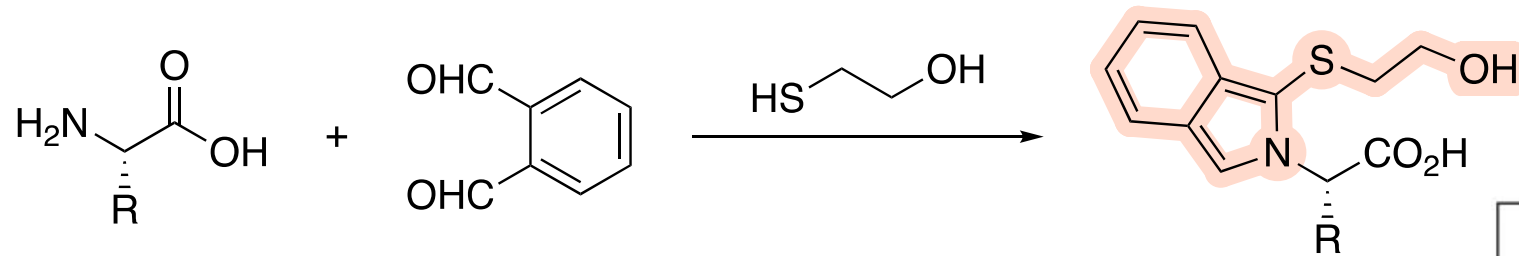
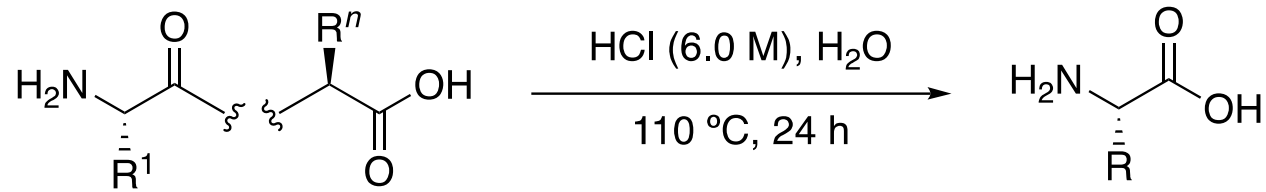
- What is the order of amino acid residues in a polypeptide chain from *N*- to *C*-terminus?



N AVSKVYARSV YDSRGNPTVE
VELTTEKGVF RSIVPS... **C** 🔗

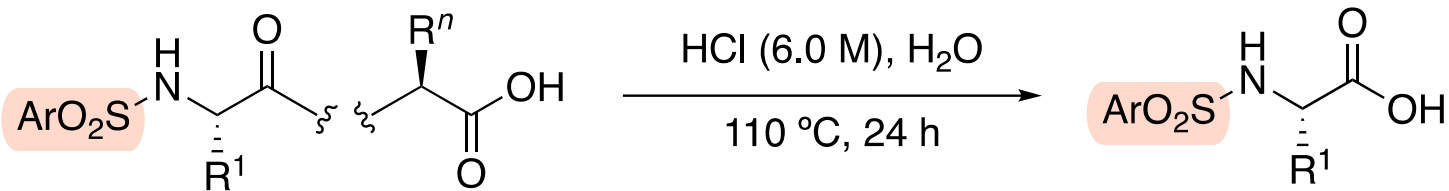
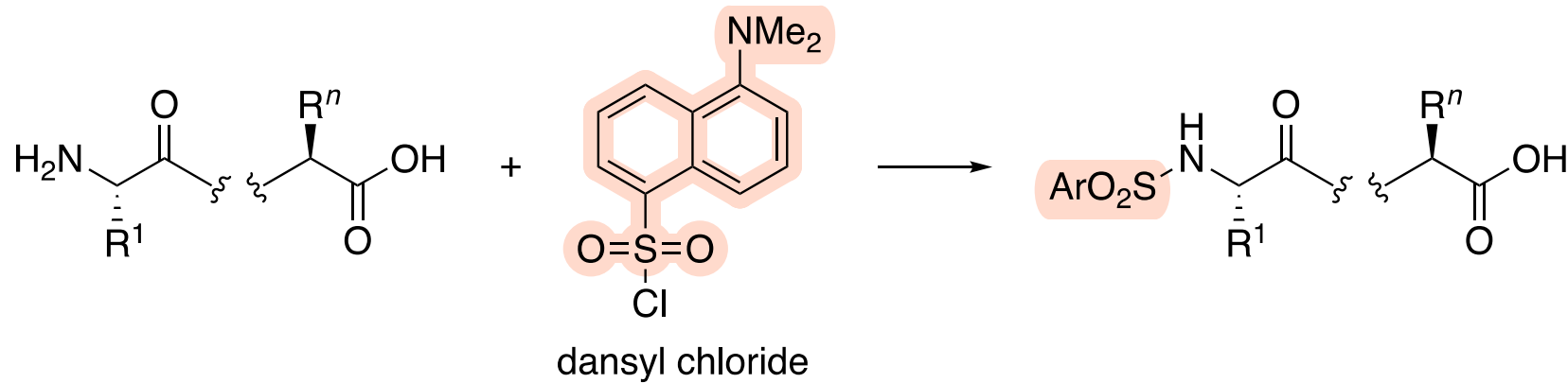
Hydrolysis and Tagging to Determine Polypeptide Composition

- Hydrolysis under strongly acidic conditions, followed by tagging with a fluorescent group and chromatography, reveals the types and relative amounts of amino acids in a polypeptide chain



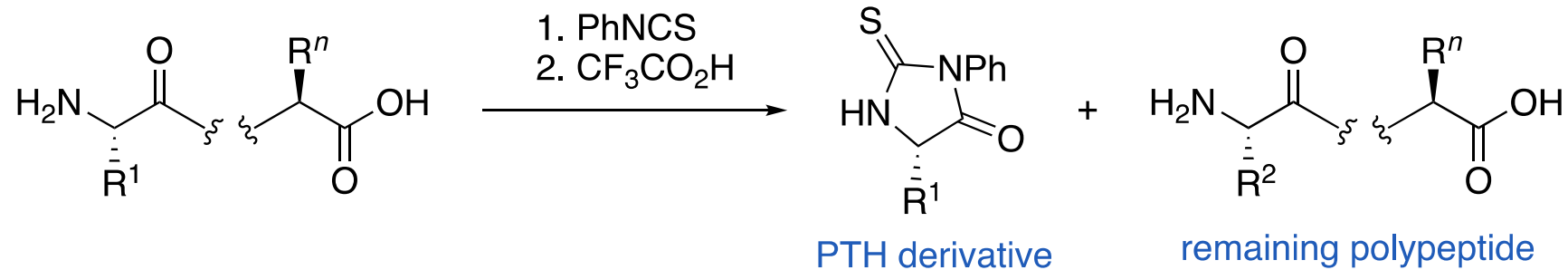
Selective Tagging of the *N*-terminus

- Using a sulfonyl chloride bearing a fluorescent group such as dansyl chloride, the *N*-terminal residue can be selectively tagged and identified



25.5 Edman Degradation for Polypeptide Sequencing | 1 of 2

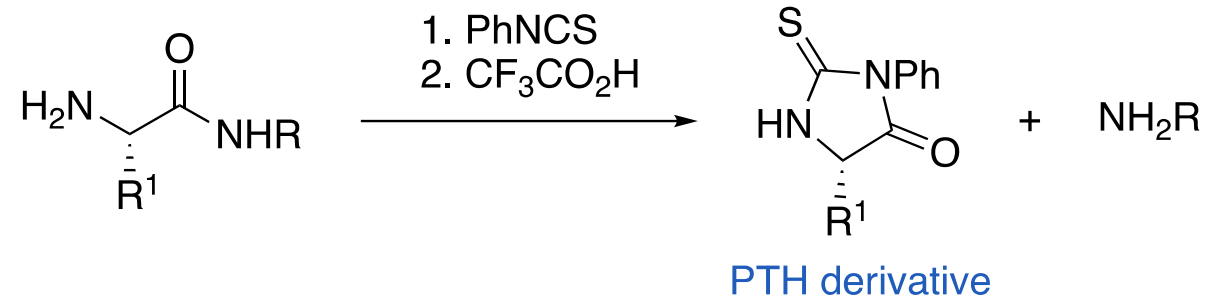
- Selective reactivity of the *N*-terminal amino group is also the basis for the sequential degradation of a polypeptide chain known as the **Edman degradation**
- Treatment of the polypeptide with phenyl isothiocyanate (PhNCS, PIC) and then trifluoroacetic acid (CF₃CO₂H, TFA) liberates the *N*-terminal amino acid residue as a phenylthiohydantoin (PTH) derivative



- The identity of the PTH derivative can be determined spectroscopically
- The process is iterated to identify each amino acid residue from 1 to *n*

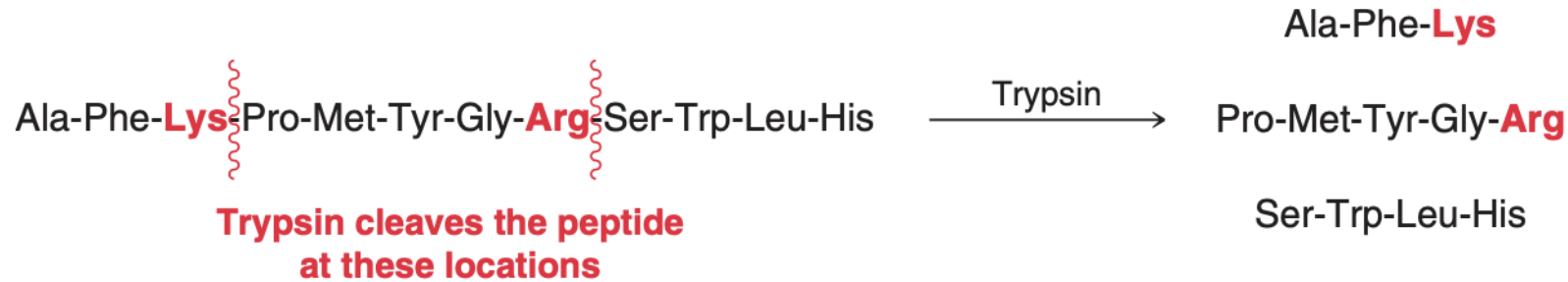
N AVSKVYARSV YDSRGNPTVE
VELTTEKGVF RSIVPS... **C**

- The mechanism of *N*-terminal cleavage involves addition, substitution, and rearrangement

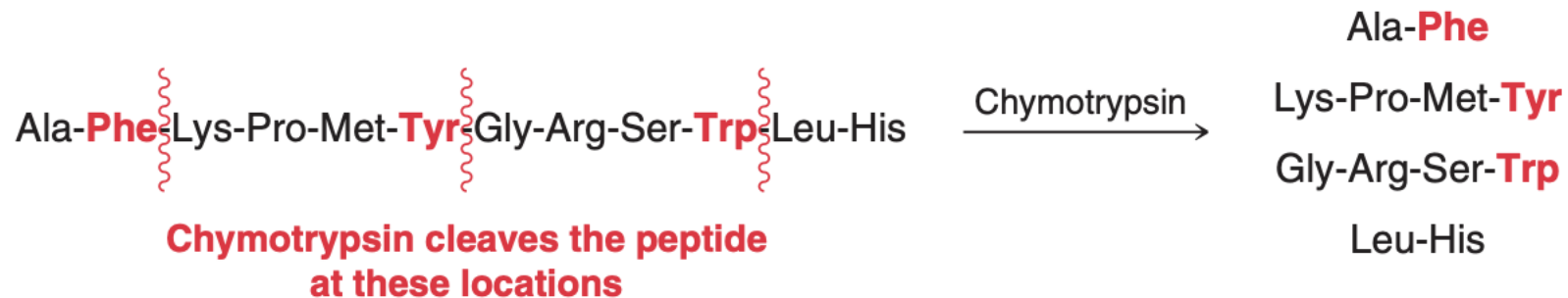


25.5 Peptidases Cleave Long Polypeptide Chains into Smaller Fragments

- Peptidase (or protease) enzymes cleave polypeptide chains at specific residues
- **Trypsin** cleaves at the C-terminal side of arginine and lysine (basic) residues



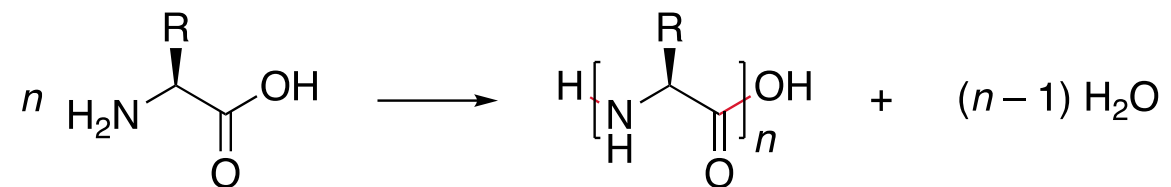
- **Chymotrypsin** cleaves at the C-terminal side of phenylalanine, tyrosine, and tryptophan (aromatic) residues



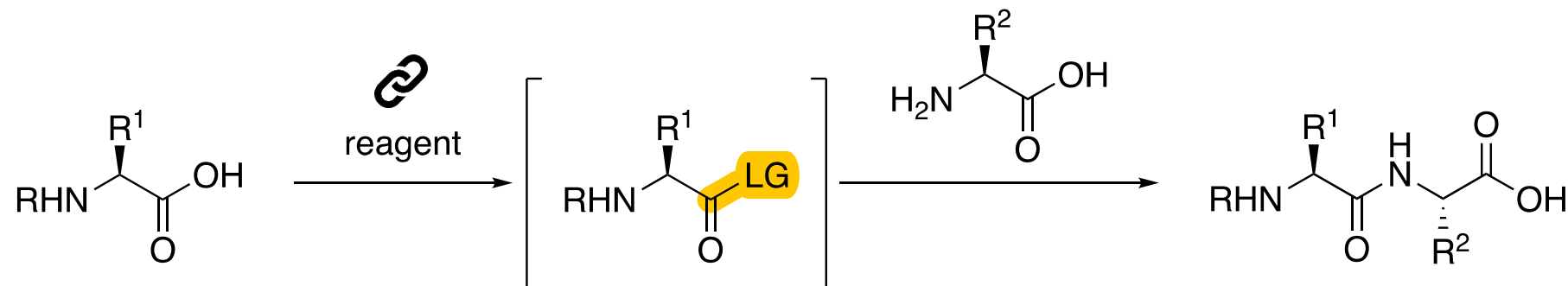
- Both sets of results are necessary to deduce the sequence of the full polypeptide

25.6 Polypeptide Synthesis via Peptide Coupling

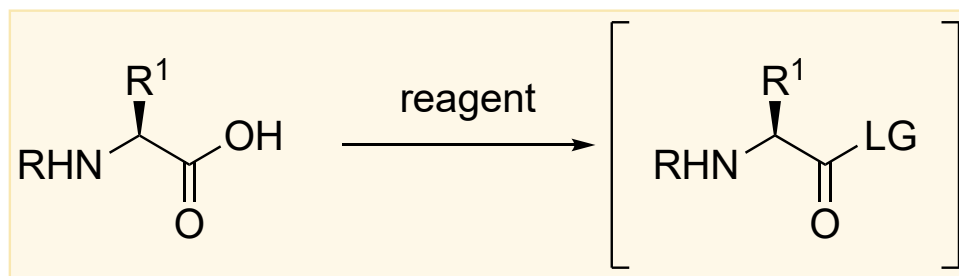
- Synthesis of a polypeptide from amino acids involves repeated nucleophilic acyl substitution (S_NAc) or **peptide coupling**



- However, hydroxyl is a poor leaving group; the carboxylic acid must be *activated* by converting $-\text{OH}$ to a better leaving group using a **peptide coupling reagent**

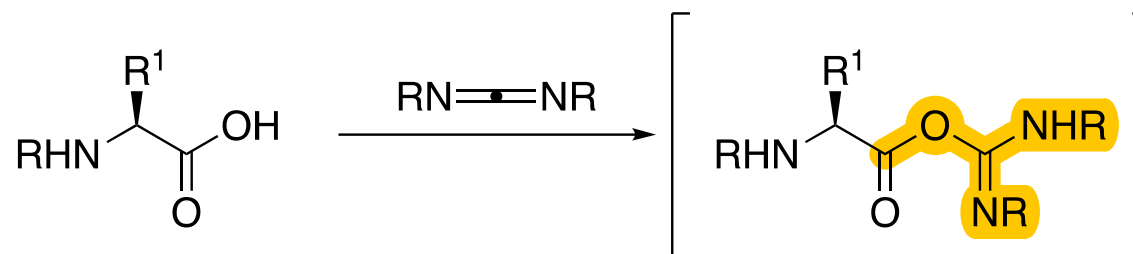


25.6 Peptide Coupling Reagents



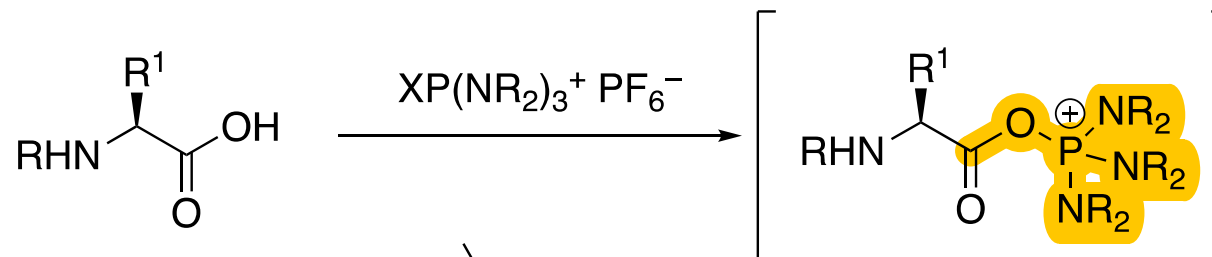
- Carbodiimides

- Dicyclohexylcarbodiimide (DCC)
- Diisopropylcarbodiimide (DIC)



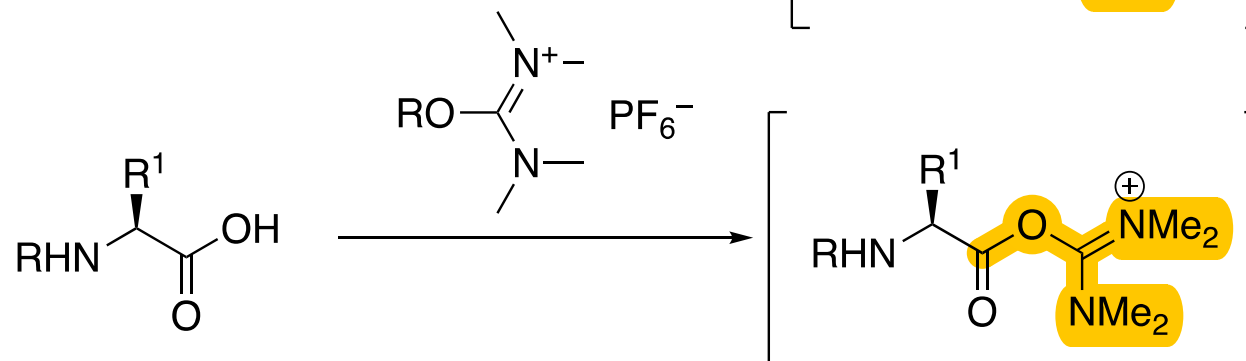
- Phosphonium salts

- BOP
- PyBOP
- PyBrOP



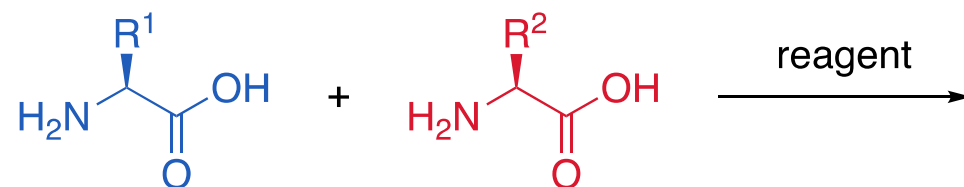
- Uronium salts

- HATU
- HBTU

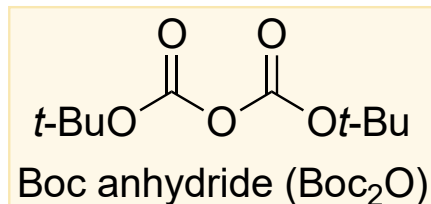
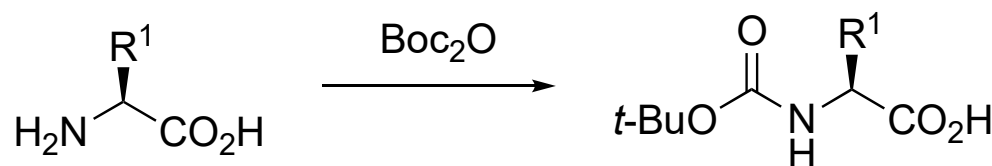


25.6 The Problem of Polypeptide Synthesis

- Each amino acid includes a nucleophilic amino group and electrophilic carbonyl group, so we must *block* one amino group and *activate* one carbonyl group to achieve selective reaction

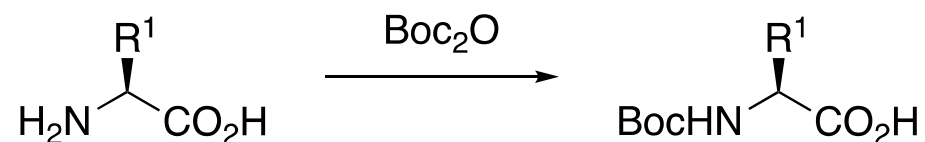


- To protect the *N*-terminal amino group, we place it in conjugation with a carbonyl group

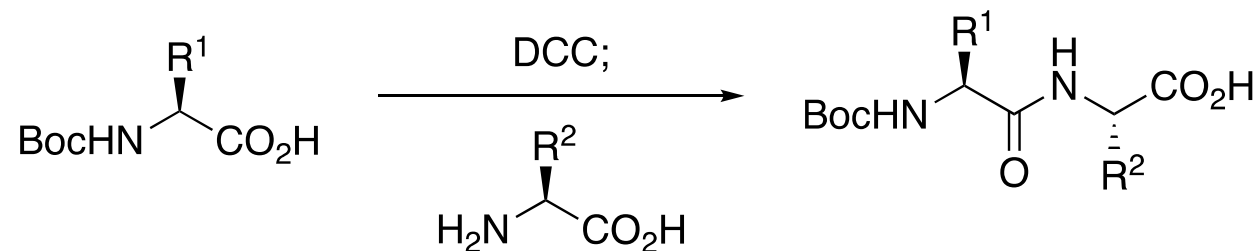


- Polypeptides are synthesized from *N*- to *C*-terminus using an iterative process

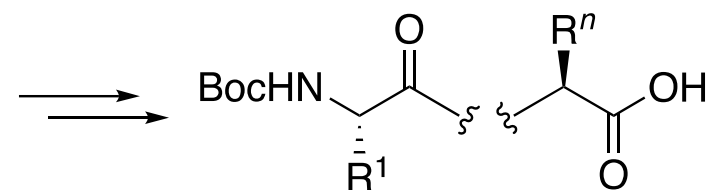
1. Protect or block the *N*-terminal amino group. (Optionally, protect the *C*-terminal carboxyl group as an ester.)



2. (Optionally, deprotect the *C*-terminal carboxyl group.) Activate the carboxyl group with a coupling reagent.
3. Add the nucleophilic amino acid. (Optionally, add as an ester.)

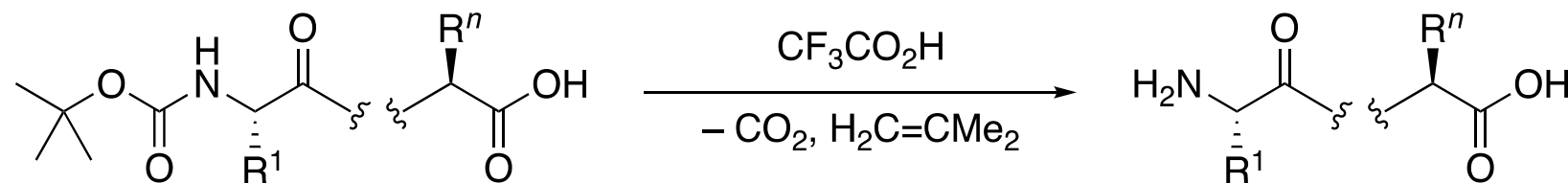


4. Repeat steps 2 and 3 until the *C*-terminus is reached.

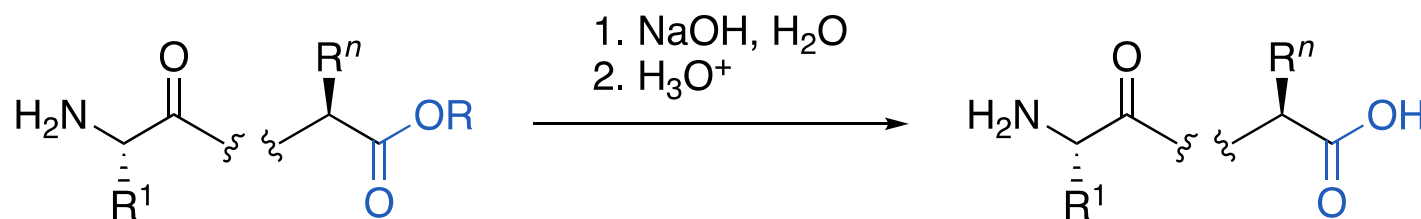


25.6 Polypeptide Synthesis: The Big Idea | 2 of 2

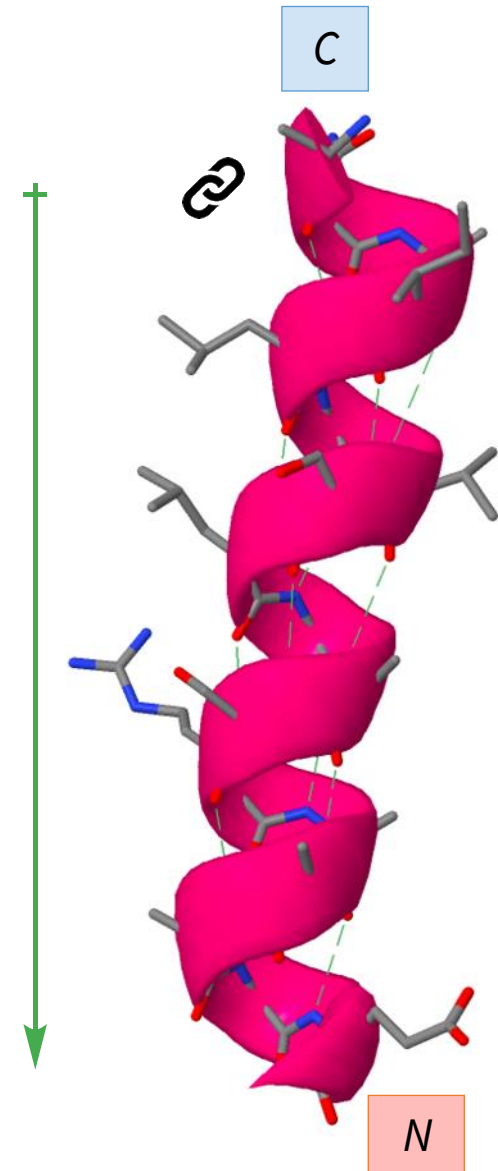
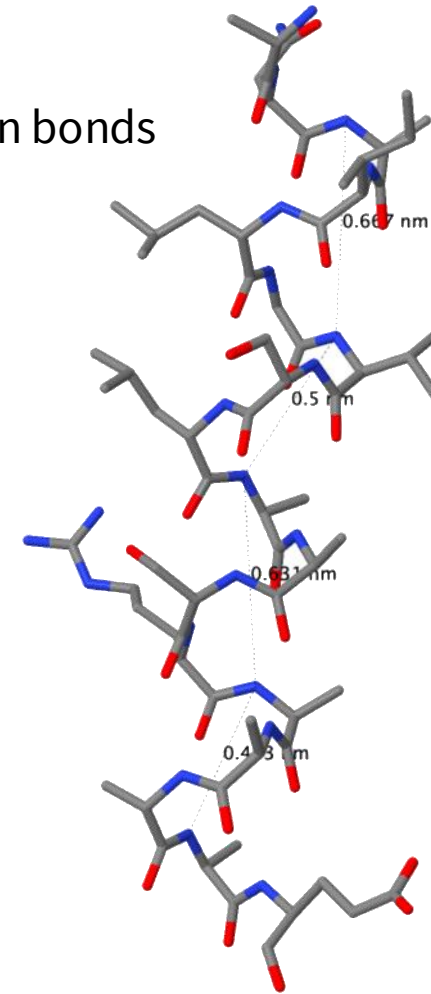
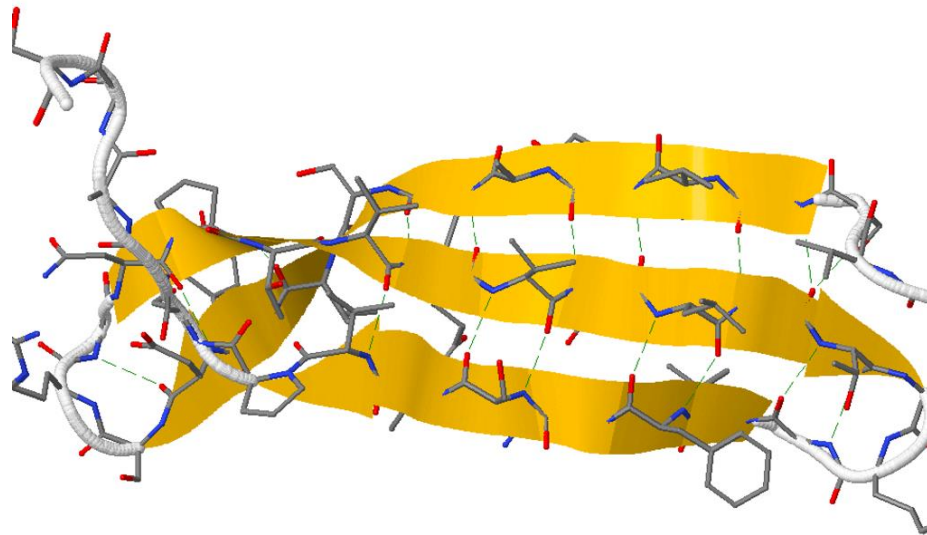
- The *N*-terminal Boc group is cleaved using trifluoroacetic acid (TFA, $\text{CF}_3\text{CO}_2\text{H}$)
- Mechanism involves proton transfer followed by elimination of gaseous carbon dioxide and isobutylene



- If amino acid esters are used, the C-terminal carboxyl group is deprotected using aqueous sodium hydroxide



- **Primary structure** is the sequence of amino acid residues in a polypeptide chain
- **Secondary structures** are larger-scale structures spanning multiple residues that are held together by noncovalent interactions
 - **α -Helices** are helical structures held together by lateral hydrogen bonds roughly parallel to the axis of the helix (usually right handed)
 - **β -Pleated sheets** are composed of **β -strands**, roughly flat strands of amino acid residues with side chains projecting above and below



- **Tertiary structure** refers to the three-dimensional shape of the polypeptide chain, including recurring collections of secondary structures called **domains**
 - Divided into all-alpha, all-beta, and alpha/beta domains
 - Databases such as the Structural Classification of Proteins (SCOP) database organize and classify proteins by their domains
- Nonpolar side chains are typically found in the core of proteins while polar side chains are typically found on the surface
- **Quaternary structure** arises in a protein consisting of multiple polypeptide chains

